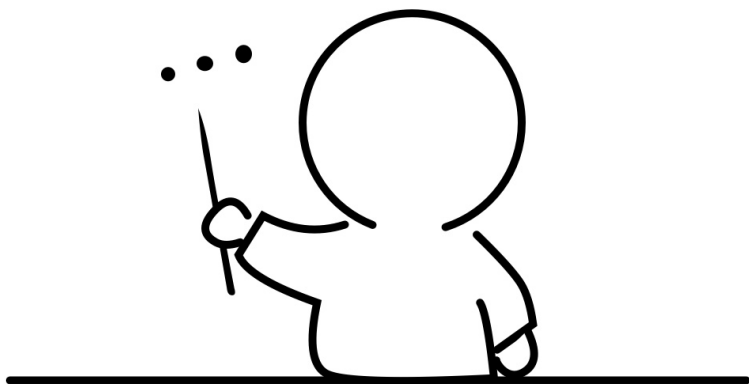


It's time for class

Daily Reminder



beautiful things happen
when you do the work to reprogram
that negative voice in your head

HW08

1. 不成立

下举反例: $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ $A+B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

$$\det A = -1 \quad \det B = 2 \quad \det(A+B) = 3$$

$$\det(A+B) \neq \det A + \det B$$

$$\text{证: } \det(A+B) = \det(a_1+b_1, a_2+b_2, \dots, a_n+b_n)$$

$$= \sum \det(x_1, x_2, \dots, x_n)$$

$x = a$ 或 b 且满足 $\sum$ 求和中两两

数学形式不同

而 $\det A + \det B$

$$= \det(a_1, a_2, \dots, a_n) + \det(b_1, b_2, \dots, b_n)$$

二者并不相同

2. 不成立.

$$\det(\lambda A) = \det(\lambda a_1, \lambda a_2, \dots, \lambda a_n)$$

$$= \lambda^n \det(a_1, a_2, \dots, a_n) = \lambda^n \det A$$

$$\text{又 } \lambda \notin \{0, 1\} \quad \therefore \det(\lambda A) \neq \lambda \det A$$

3. 利用Gram-Smith正交化将 $v_1 \sim v_k$ 化为一组长度为1且彼此正交的向量 $u_1 \sim u_k$

$$v_i = \sum_{j=1}^i c_{ij} u_j \quad (c_{ij})_{k \times k} \text{ 是下三角矩阵}$$

$$\begin{aligned} \langle v_i, v_j \rangle &= \left\langle \sum_{m=1}^i c_{im} u_m, \sum_{l=1}^j c_{jl} u_l \right\rangle \\ &= \sum_{m=1}^i \sum_{l=1}^j c_{im} c_{jl} \langle u_m, u_l \rangle \\ &= \sum_{m=1}^k c_{im} c_{jm} \quad (c_{ij} \text{ 是下三角矩阵}) \end{aligned}$$

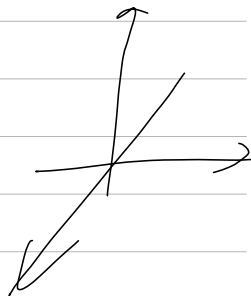
$$\begin{aligned} \text{记 } G &= (\langle v_i, v_j \rangle)_{k \times k} \\ &= \left(\sum_{k=1}^{\min\{i,j\}} c_{ik} c_{jk} \right)_{k \times k} \end{aligned}$$

$$\text{设 } A = (v_1 \dots v_k)$$

$$v_1 = c_{11} u_1$$

$$v_2 = c_{21} u_1 + c_{22} u_2$$

$\vdots$



$$\begin{aligned} G &= \begin{bmatrix} c_{11} & & & \\ c_{21} & c_{22} & & \\ \vdots & & \ddots & \\ c_{k1} & \dots & \dots & c_{kk} \end{bmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_k \end{pmatrix} (u_1 \dots u_k) \cdot \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1k} \\ & c_{22} & & \\ & & \ddots & \\ & & & c_{kk} \end{bmatrix} \\ &= \begin{bmatrix} c_{11} & & & \\ c_{21} & c_{22} & & \\ \vdots & & \ddots & \\ c_{k1} & \dots & \dots & c_{kk} \end{bmatrix} \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1k} \\ & c_{22} & & \\ & & \ddots & \\ & & & c_{kk} \end{bmatrix} = A^T A \end{aligned}$$

$$\det G = (c_{11} c_{22} \dots c_{kk})^2 \quad \text{另解: 用施密特正交化作为非单位的基 (不需要单位化)}$$

$$\sqrt{\det G} = |c_{11} c_{22} \dots c_{kk}|$$

$$\therefore c_{11} = \frac{\|v_1\|^2}{\|u_1\|^2}$$

$$c_{22} = \frac{\|v_2 - c_{21} u_1\|^2}{\|u_2\|^2}$$

$$c_{22} = \frac{\|v_2 - c_{21} u_1\|^2}{\|u_2\|^2}$$

$$c_{33} = \frac{\|v_3 - c_{31} u_1 - c_{32} u_2\|^2}{\|u_3\|^2}$$

⋮

$$\therefore c_{11} c_{22} \dots c_{kk} = V$$

$$\therefore V = \sqrt{\det (\langle v_i, v_j \rangle)_{k \times k}}$$

$v_1 \dots v_k$ 线性无关
 $u_1 \dots u_k$ 线性无关

$$[u_1 \dots u_k] = [v_1 \dots v_k] \begin{bmatrix} 1 & x & x & x \\ & 1 & x & x \\ & & \ddots & x \\ & & & 1 \end{bmatrix} \quad \begin{matrix} \uparrow \\ M \end{matrix}$$

$$G_2 = M^T G_1 M$$

(证明: 看 i, j 元)

$$|G_2| = |M^T| \cdot |G_1| \cdot |M| = |G_1|$$

4.

$$Av_j = \sum_{i=1}^n a_{ij} v_i$$

$$(Av_1 \dots Av_n) = (v_1 \dots v_n) \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & & & \vdots \\ a_{n1} & \dots & \dots & a_{nn} \end{pmatrix}$$

两组基分别为 $[b_i]_B$, $[b_i]_C$

线性表示

$$(Av_1 \dots Av_n) = (v_1 \dots v_n) M_A$$

$$[b_i]_C = P_{B \rightarrow C} [b_i]_B$$

$$[Abi]_C = [bi]_C \cdot M_C$$

$$[Abi]_B = [bi]_B \cdot M_B$$

$$[bi]_C M_C = [Abi]_C = [A(P_{B \rightarrow C} bi)]_B$$

$$= P_{B \rightarrow C} [A bi]_B = P_{B \rightarrow C} [bi]_B M_B = [bi]_C M_B$$

$$\text{记 } [bi]_C = [c_1 \ c_2 \ \dots \ c_n]$$

$$\det [c_1 \ c_2 \ \dots \ c_n] \cdot \det M_B = \det [c_1 \ c_2 \ \dots \ c_n] \det M_C$$

$\therefore c_1 \dots c_n$ 线性无关

$$\therefore \det [c_1 \dots c_n] \neq 0$$

$$\therefore \det M_B = \det M_C$$

另解：设另一组基为 $u_1 \dots u_n$

$$(Av_1 \dots Av_n) = (v_1 \dots v_n) M_A$$

$$(Au_1 \dots Au_n) = (u_1 \dots u_n) M_B$$

$$\text{设 } (v_1 \dots v_n) = (u_1 \dots u_n) T$$

$$(Av_1 \dots Av_n) = (Au_1 \dots Au_n) T$$

$$\text{记：} \because Av_1 = A(k_1 u_1 + \dots + k_n u_n) = k_1 Au_1 + \dots + k_n Au_n$$

$$\therefore (Au_1 \dots Au_n) T = (v_1 \dots v_n) M_A$$

$$(u_1 \dots u_n) M_B T = (u_1 \dots u_n) T M_A$$

$$M_B T = T M_A \quad (M_B) = (M_A)$$

5 至少需要做五次行星位置的观测
 $P(x, y)$ 位于五点 $P_k(x_k, y_k) (k=1, 2, 3, 4, 5)$
 所确定的圆锥曲线上当且仅当

$$(\alpha x^2 + \beta x y + \gamma y^2 + \delta x + \epsilon y + \zeta = 0)$$

$$\begin{bmatrix} x^2 & xy & y^2 & x & y & 1 \\ x_1^2 & x_1 y_1 & y_1^2 & x_1 & y_1 & 1 \\ x_2^2 & x_2 y_2 & y_2^2 & x_2 & y_2 & 1 \\ x_3^2 & x_3 y_3 & y_3^2 & x_3 & y_3 & 1 \\ x_4^2 & x_4 y_4 & y_4^2 & x_4 & y_4 & 1 \\ x_5^2 & x_5 y_5 & y_5^2 & x_5 & y_5 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ \beta \\ \gamma \\ \delta \end{bmatrix} = 0$$

$$A\vec{x} = 0 \text{ 有唯一解}$$

$$|\ker(A)| = 1$$

$$b = \dim \ker(A) + \text{rank}(A)$$

$$\text{rank}(A) = 5$$

这是 $k \times 6$ 矩阵秩为 5
 $k > 5$ $\therefore$ 至少 5 次

有解

$\therefore$ 方程为

$$\begin{vmatrix} x^2 & xy & y^2 & x & y & 1 \\ x_1^2 & x_1 y_1 & y_1^2 & x_1 & y_1 & 1 \\ x_2^2 & x_2 y_2 & y_2^2 & x_2 & y_2 & 1 \\ x_3^2 & x_3 y_3 & y_3^2 & x_3 & y_3 & 1 \\ x_4^2 & x_4 y_4 & y_4^2 & x_4 & y_4 & 1 \\ x_5^2 & x_5 y_5 & y_5^2 & x_5 & y_5 & 1 \end{vmatrix} = 0$$

圆锥曲线需满足的条件:

$$\begin{vmatrix} x_1^2 & x_1 y_1 & y_1^2 & x_1 & y_1 \\ x_2^2 & x_2 y_2 & y_2^2 & x_2 & y_2 \\ x_3^2 & x_3 y_3 & y_3^2 & x_3 & y_3 \\ x_4^2 & x_4 y_4 & y_4^2 & x_4 & y_4 \\ x_5^2 & x_5 y_5 & y_5^2 & x_5 & y_5 \end{vmatrix} \neq 0$$

或:

$$\begin{bmatrix} x_1^2 & \cdots & y_1 \\ \vdots & & \vdots \\ x_5^2 & \cdots & y_5 \end{bmatrix} \begin{pmatrix} a \\ b \\ c \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} -\delta \\ -\epsilon \\ -\zeta \\ -\delta \\ -\epsilon \end{pmatrix}$$

有唯一解 $\therefore |A| \neq 0$

6 值为 $\prod_{1 \leq j < i \leq n} (x_i - x_j)$, 记为 $D(n)$

证明: $1 \leq j < i \leq n$

$$D(n) = \begin{vmatrix} 1 & x_1 & x_1^2 & \dots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \dots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^{n-1} \end{vmatrix} = \begin{vmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \\ x_1^{n-2} & x_2^{n-2} & \dots & x_n^{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ x_1^{n-1} & x_2^{n-1} & \dots & x_n^{n-1} \end{vmatrix} \quad \begin{matrix} \text{(从第 } n \text{ 行开始} \\ \text{每一行 - 前一行} \\ \text{ } \times 1 \text{ 倍)} \end{matrix}$$

$$= \begin{vmatrix} 1 & 1 & \dots & 1 \\ 0 & x_2 - x_1 & \dots & x_n - x_1 \\ 0 & x_2^2 - x_1 x_2 & \dots & x_n^2 - x_1 x_n \\ \vdots & \vdots & \ddots & \vdots \\ 0 & x_2^{n-1} - x_1 x_2^{n-2} & \dots & x_n^{n-1} - x_1 x_n^{n-2} \end{vmatrix} = \begin{vmatrix} x_2 - x_1 & \dots & x_n - x_1 \\ x_2^2 - x_1 x_2 & \dots & x_n^2 - x_1 x_n \\ \vdots & \ddots & \vdots \\ x_2^{n-1} - x_1 x_2^{n-2} & \dots & x_n^{n-1} - x_1 x_n^{n-2} \end{vmatrix}$$

$$= (x_2 - x_1)(x_3 - x_1) \dots (x_n - x_1) \cdot \begin{vmatrix} 1 & 1 & \dots & 1 \\ x_2 & \dots & x_n \\ \vdots & \ddots & \vdots \\ x_2^{n-2} & \dots & x_n^{n-2} \end{vmatrix}$$

$$D(n) = (x_2 - x_1) \dots (x_n - x_1) \cdot D(n-1)$$

$$\therefore D(n) = \prod_{1 \leq j < i \leq n} (x_i - x_j)$$

或用第 k 行 - 若干倍行, 逐步往下消

另解:

$$n=2 \quad \begin{vmatrix} 1 & 1 \\ x_1 & x_2 \end{vmatrix} = x_2 - x_1$$

$$n=3 \quad \begin{vmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \end{vmatrix} = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2)$$

猜想 (n) $|A_n| = \prod_{1 \leq j < i \leq n} (x_i - x_j)$

用归纳法

假设 $n-1$ 成立

$|A_n|$ 进行化简,

$$|A_n| = (x_n - x_{n-1})(x_n - x_{n-2}) \cdots (x_n - x_1) \cdot |A_{n-1}|$$

$$= \prod_{1 \leq j < i \leq n} (x_i - x_j)$$

A_n 也成立

7. 存在性: 思路是找到一个在 x_i 处取值为 1,
 x_i 处取值为 0 的多项式 $l_j(x)$

$$f = \sum_{j=1}^{n+1} y_j l_j(x)$$

$$\text{此时 } f(x_k) = \sum_{j=1}^{n+1} y_j l_j(x_k) = y_k$$

$$l_j(x) = \frac{\prod_{i \neq j} (x - x_i)}{\prod_{i \neq j} (x_j - x_i)} \quad \text{次数不超过 } n$$

另解:

$$f(x_i) = y_i = a_0 + a_1 x_i + a_2 x_i^2 + \dots + a_n x_i^n$$

$$a_0 + a_1 x_i + a_2 x_i^2 + \dots + a_n x_i^n = y_i$$

$$\begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^n \\ 1 & x_2 & \dots & x_2^n \\ \vdots & \vdots & & & \vdots \\ 1 & x_{n+1} & \dots & x_{n+1}^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

由 Cram 法则
 (第 i 个)

$$a_{i-1} =$$

$$\frac{\det(a_1 \ a_2 \ \dots \ b \ \dots \ a_n)}{\det(a_1 \ \dots \ a_n)} \quad \text{或: } \det(A) \neq 0 \quad \therefore \text{有唯解}$$

$$\frac{\det(a_1 \ a_2 \ \dots \ b \ \dots \ a_n)}{\det(a_1 \ a_2 \ \dots \ a_n)} \quad \therefore X = A^{-1}b$$

$$= \frac{1}{\prod_{i>j} (x_i - x_j)}$$

唯一性:

假设存在两个这样的多项式 f_1, f_2

$$\text{使 } y_k = f_1(x_k) = f_2(x_k)$$

$$(f_1 - f_2)(x_k) = 0 \quad k = (1, 2, \dots, n+1)$$

$$\therefore f_1 - f_2 = \lambda \cdot (x - x_1)(x - x_2) \dots (x - x_{n+1})$$

若 $\lambda \neq 0$, $f_1 - f_2$ 是 $n+1$ 次

但 f_1 和 f_2 都是 n 次, 矛盾

$$\therefore \lambda = 0 \quad \text{即 } f_1 = f_2$$

$$7. p(x) = \sum_{k=1}^m \sum_{j=0}^{n_k} y_k^{(j)} h_{k,j}(x)$$

其中 $h_{k,j}(x)$ 是形如 $(x - x_k)^j$ 的基函数,

满足在 $x = x_k$ 处第 j 阶导数为 1,

在其他插值点和其它阶导数上为 0

8. 设 $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ $y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ (阶为1)

$$xy^T = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} [y_1 \dots y_n] = \begin{bmatrix} x_1 y_1 & x_1 y_2 & \dots & x_1 y_n \\ x_2 y_1 & x_2 y_2 & \dots & x_2 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_n y_1 & x_n y_2 & \dots & x_n y_n \end{bmatrix}$$

$$|I + xy^T| = \begin{vmatrix} 1+x_1 y_1 & x_1 y_2 & \dots & x_1 y_n \\ x_2 y_1 & 1+x_2 y_2 & \dots & x_2 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_n y_1 & \dots & \dots & 1+x_n y_n \end{vmatrix}$$

$$= (1+x_1 y_1) \cdot \begin{vmatrix} 1+x_2 y_2 - \frac{x_1 x_2 y_1 y_2}{1+x_1 y_1} & \dots & x_2 y_n - \frac{x_1 x_2 y_1 y_n}{1+x_1 y_1} \\ \vdots & \ddots & \vdots \\ x_n y_2 - \frac{x_1 x_n y_1 y_2}{1+x_1 y_1} & \dots & 1+x_n y_n - \frac{x_1 x_n y_1 y_n}{1+x_1 y_1} \end{vmatrix}$$

$$= (1+x_1 y_1 + x_2 y_2) \cdot \begin{vmatrix} \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \end{vmatrix}$$

逐步展开, 结果为

$$1 + x^T y$$

另解:

$$\begin{vmatrix} 1+x_1 y_1 & x_1 y_2 & \dots & x_1 y_n \\ x_2 y_1 & 1+x_2 y_2 & \dots & x_2 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_n y_1 & \dots & \dots & 1+x_n y_n \end{vmatrix} = \begin{vmatrix} 1 & x_1 y_2 & \dots & x_1 y_n \\ 0 & \vdots & \ddots & \vdots \\ 0 & \vdots & \ddots & \vdots \end{vmatrix} + \begin{vmatrix} x_1 y_1 & x_1 y_2 & \dots & x_1 y_n \\ x_2 y_1 & \vdots & \ddots & \vdots \\ x_n y_1 & \dots & \dots & \vdots \end{vmatrix}$$

$$= \begin{vmatrix} 1+x_2 y_2 & \dots & x_2 y_n \\ \vdots & \ddots & \vdots \end{vmatrix} + x_1 y_1 \cdot \begin{vmatrix} 1+x_2 y_2 & \dots & x_2 y_n \\ \vdots & \ddots & \vdots \end{vmatrix} \quad \text{以此逐步展开即可}$$

9. 几何解释: 当列向量线性无关时 $|\det(A)|$ 表示它们构成的 n 维多面体的体积, 一定小于等于各长度的乘积
当线性相关, $|\det A| = 0$. 等式同样成立

通过施密特正交化

可将 A 的列向量 $v_1 \dots v_n$ 转化为一组正交基 $u_1 \dots u_n$

$$|\det A| = |\det U| = \prod_{i=1}^n \|u_i\|$$

$$\text{又 } \|u_i\| \leq \|v_i\|$$

$$\therefore |\det A| \leq \prod_{i=1}^n \|v_i\|$$

第四题已证

$$10. p_n = b p_{n-1} - a c p_{n-2}$$

$$p_1 = b \quad p_2 = b^2 - ac \quad (\text{由此可推出 } p_0 = 1)$$

$$\lambda^n = b \lambda^{n-1} - a c \lambda^{n-2}$$

$$\lambda^2 - b\lambda + ac = 0$$

$$\lambda_{1,2} = \frac{b \pm \sqrt{b^2 - 4ac}}{2}$$

$$\text{不妨 } \lambda_2 > \lambda_1 \quad \lambda_2 - \lambda_1 = \sqrt{b^2 - 4ac}$$

$$p_n = A \lambda_1^n + B \lambda_2^n$$

$$\begin{cases} p_1 = A \lambda_1 + B \lambda_2 = b \\ p_0 = A + B = 1 \end{cases} \quad \begin{cases} A = \frac{\lambda_2 - b}{\lambda_2 - \lambda_1} \\ B = \frac{b - \lambda_1}{\lambda_2 - \lambda_1} \end{cases}$$

$$\therefore p_n = \frac{\sqrt{b^2 - 4ac} - b}{2\sqrt{b^2 - 4ac}} \left(\frac{b - \sqrt{b^2 - 4ac}}{2} \right)^n + \frac{b + \sqrt{b^2 - 4ac}}{2\sqrt{b^2 - 4ac}} \left(\frac{b + \sqrt{b^2 - 4ac}}{2} \right)^n$$

$$p_n = \frac{(b + \sqrt{b^2 - 4ac})^{n+1} - (b - \sqrt{b^2 - 4ac})^{n+1}}{2^{n+1} \sqrt{b^2 - 4ac}}$$